From Prephysical Hypotheses to Physical Models

Structure, observation and physical identification

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Proposed relational structures
Relations and symmetries specify a mathematical description. Its observable meaning and physical units require a justified assignment.

Representation and apparatus
Specify states, operators, preparations, accessible effects and units. Retain uncertain detector, source and clock calibrations in the model class.

Which properties are shared by every description with the same outcome laws?

Specified observations

Conditional inferences and their limits

Algebra and symmetry
Known finite operators, their commutant, and spin fluctuations in a specified state.

Commutant multiplicities control invariant rank-one choices. Vanishing averaged commutators alone do not identify independent subsystems.

Labelled spectral probes
Matched populations, known ports, zero-mode probabilities and unconditional first scale moments.

Nonzero zero-mode contrast on every labelled pair recovers the graph generator in the given units. Imaginary coherences can conceal it.

States, spectra and dynamics
Equal-time fermionic covariances, positive spectral moments and calibrated dynamical measurements.

Distinct dispersions can share an occupied projector. Moments constrain support only in the stated measure class. Wilson entropy depends on the mass-scaling regime.

Phase and purity
A known match population and a common phase detector with nonzero overlap on both probed directions.

A saturating harmonic fixes a rank-one block at positive weight. A detector at an overlap endpoint can leave the coherence undetermined.

Four-input quantum instrument
All four success rates and the specified binary records, with one common first-qubit detector and the stated product structure.

For the stated example, calibration narrows the compatible local Choi-weight interval from $[0, 0.9907077\ldots]$ to $[0, 37/38]$. This does not identify a unique operation.

What would test the physical interpretation?

Independent controls and observations reserved for testing
Anchor preparations, port bases, units and readout effects. Vary source populations, losses and detector orientation. Fix eligibility before private randomization and retain failed preparations. Include sampling and calibration uncertainty jointly.

Compatible descriptions
Report the surviving class or property interval. Seek an additional observation that separates members of that class.

Outside the admissible observation set
The test challenges the joint physical model and calibration assumptions at its stated error level.

Meaning of the arrows. The dashed arrow marks an open physical correspondence. Solid horizontal arrows connect specified observations to conditional mathematical results. The lower branches are possible conclusions of a prospective test, not reported experimental outcomes. The five examples illustrate observation problems; they do not exhaust the program or constitute a derivation of a unique physical ontology.

Sources. K. Justus, *Identification from Restricted Measurements*; *Operational Identification and Conditional Quantum Instruments*; *Finite Part of Wilson–Dirac Relative Entropy*; and *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals* (2026), in the accompanying *Mathematical Supplement*.